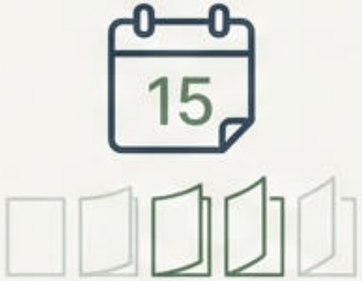
Variables in Scientific Research: The Foundation

All scientific research involves the manipulation and/or measurement of specific variables. A variable is defined as a characteristic of a person, place, or thing that can change (vary) over time or from one situation to another. Variables are fundamental to research because they represent the measurable attributes that are examined.

For example, temperature is a variable as it varies from day to day, season to season, and place to place. Similarly, height and weight are variables, as people differ in size and shape. An individual's solits, **as people differ in size** change. I's height changes until maturity, and weight can fluctuate continuously. Almost any characteristic that is not constant caan be considered a variable in a research context.

Examples of Variables

Temperature



Time

Weight

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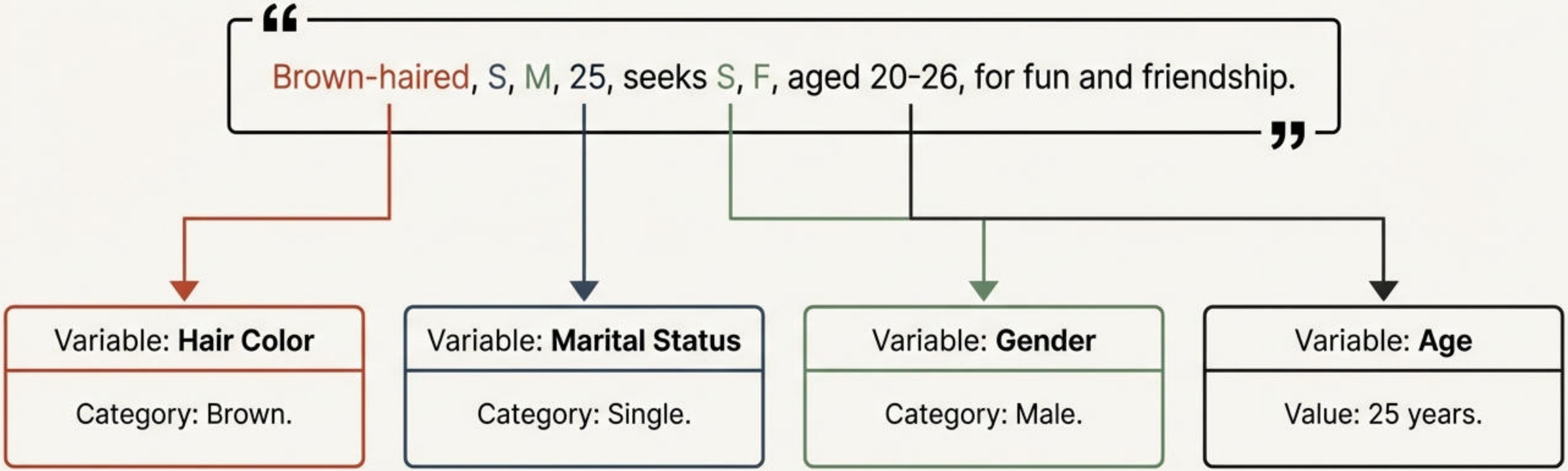
Defining and Identifying Variables in Context

Variables can be identified in everyday contexts, such as a singles advertisement: “Brown-haired, S, M, 25, seeks S, F, aged 20–26, for fun and friendship.” Each descriptor in this ad represents a variable with different potential categories or values. In this example, ‘S’ stands for ‘single,’ which is one category of the variable *marital status* (which includes categories like single, married, divorced, widowed). ‘M’ stands for “male,” which is a category of the dichotomous (two-category) variable *gender*. Age, hair color, and a preference for fun and friendship are further examples of variables present in this description.



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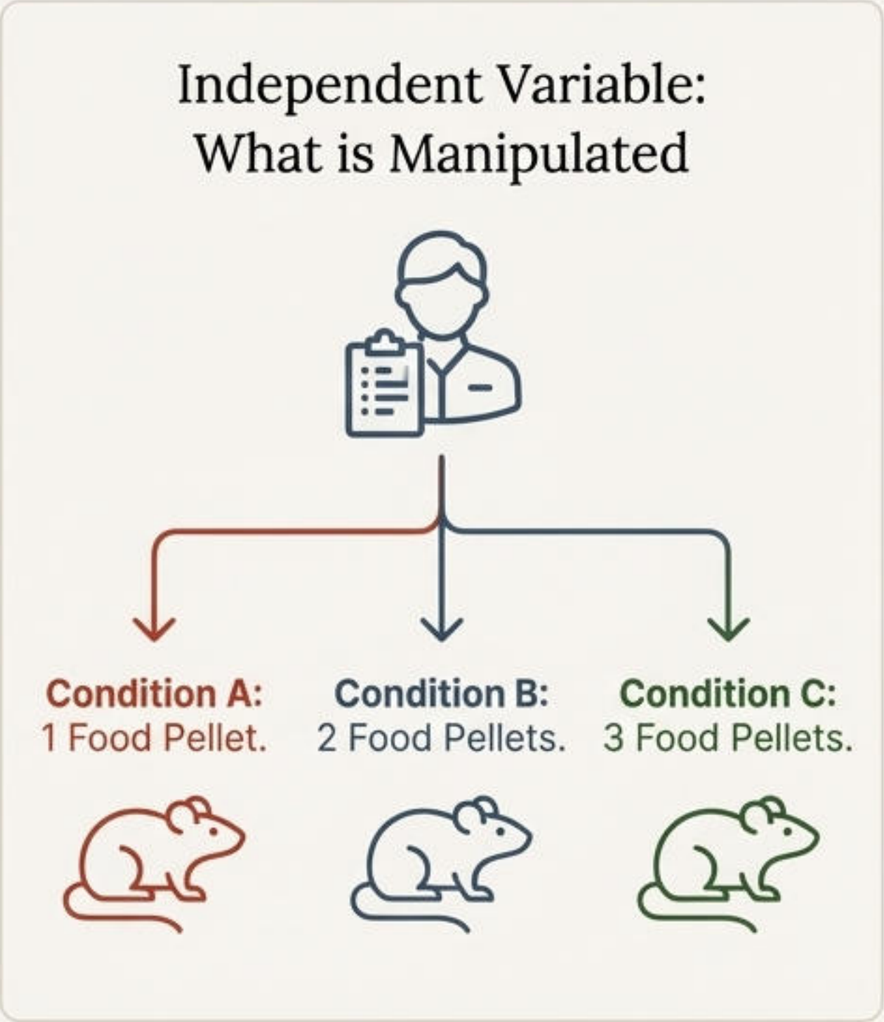
The Independent Variable: The Manipulated Factor

In an experiment, two types of variables are of primary importance. The independent variable (IV) is the aspect of an experiment that is systematically varied or manipulated by the researcher across different conditions. The purpose of this manipulation is to observe its effect on another variable.

For example, in a study on learning efficiency, researchers might investigate if the size of a reward affects maze learning in rats. The rats could be randomly assigned to different experimental conditions. In one condition, rats receive one food pellet upon reaching the goal; in another, two pellets; and in a third, three pellets. In this experiment, the independent variable is the number of food pellets, as this is the factor the researchers directly control and change for each group.



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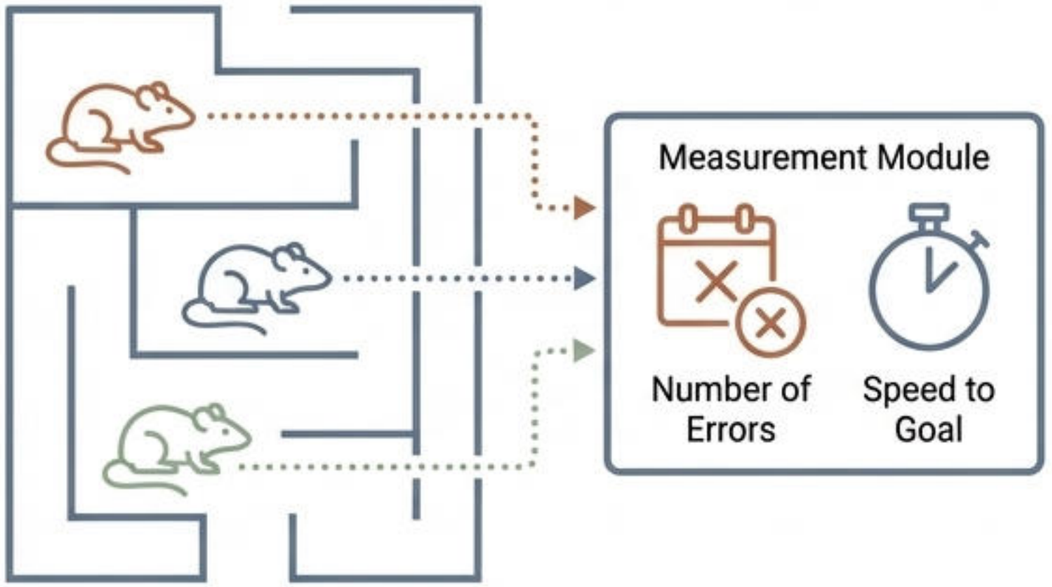
The Dependent Variable: The Measured Outcome

The dependent variable (DV) is the aspect of an experiment that is allowed to vary freely to determine if it is affected by changes in the independent variable. In other words, the dependent variable is what is measured in an experiment. In psychology, the dependent variable is always some type of behavior. Any observed changes in the dependent variable are presumed to be dependent upon the manipulation of the independent variable.

In the rat maze experiment, a possible dependent variable could be the total number of errors (wrong turns) each rat makes. Alternatively, researchers might measure the speed with which the rats reach the goal box. A significant difference in these measurements between the groups would suggest that the size of the reward (the IV) affects the efficiency of maze learning (the DV).

Dependent Variable: What is Measured



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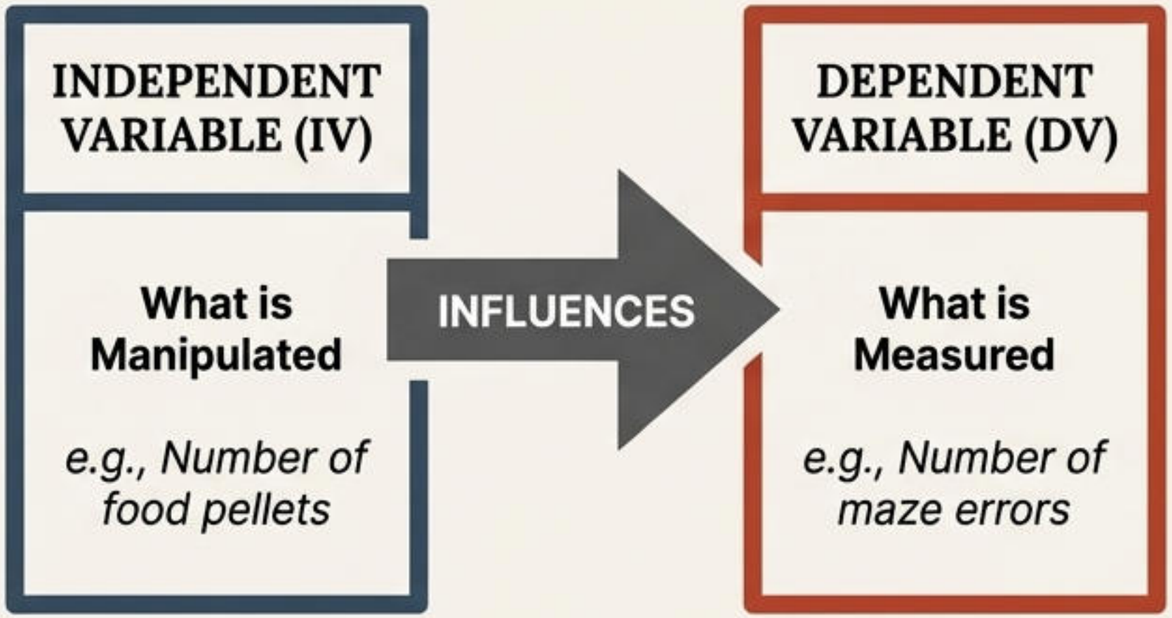
The Independent–Dependent Variable Relationship

The core of an experiment lies in the relationship between the independent and dependent variables. The independent variable is the factor manipulated by the researcher, while the dependent variable is the behavior that is measured. A useful way to distinguish between them is to remember that changes in the dependent variable are *dependent upon* changes in the independent variable.

The ultimate goal of such an experiment is to provide supportive evidence for a more general hypothesis. For example, by finding that the number of food pellets affects maze-running errors in rats, we gain evidence for the broader notion that the size of a reinforcer affects the efficiency of learning.



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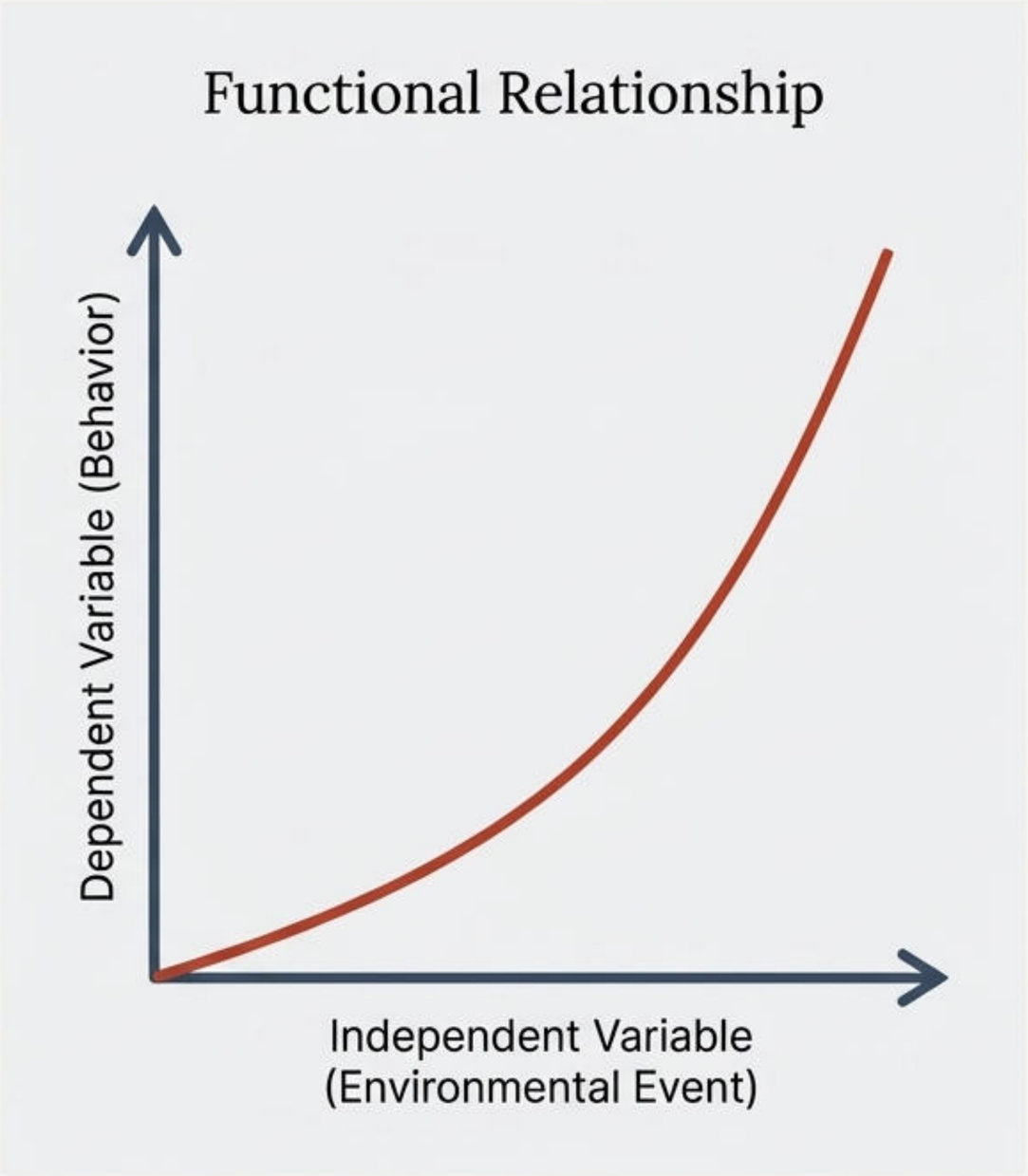
Functional Relationships in Behavioral Research

In behavioral research, the relationship between changes in an independent variable and changes in a dependent variable is known as a **functional relationship**. The independent variable is typically an environmental event, and the dependent variable is almost always a form of behavior.

Behaviorists are primarily interested in discovering these functional relationships between environmental events and behavior. This concept frames the research goal as identifying predictable, lawful connections between the environment and how an organism acts. When a change in the environment consistently produces produces a specific change in behavior, a functional relationship has been identified.



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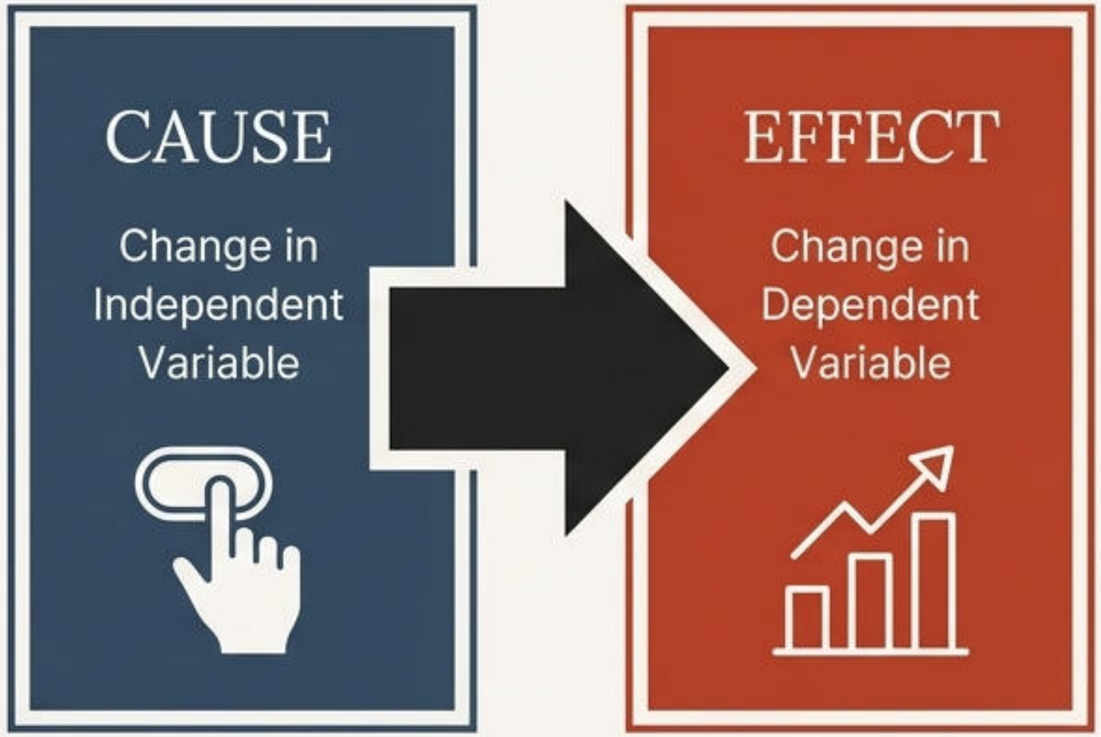
Cause and Effect in Behavioral Research

A functional relationship can be understood as a cause-and-effect relationship. In this framework, the manipulation of the independent variable is considered the cause, and the corresponding change observed in the dependent variable is the effect. For example, in a study examining the effect of viewing television violence on aggression, the level of movie violence (the independent variable) would be the cause. The children's subsequent level of aggressive behavior (the dependent variable) would be the effect. By systematically changing the cause, researchers can observe and measure its impact on the effect, thereby establishing a causal link between the two.



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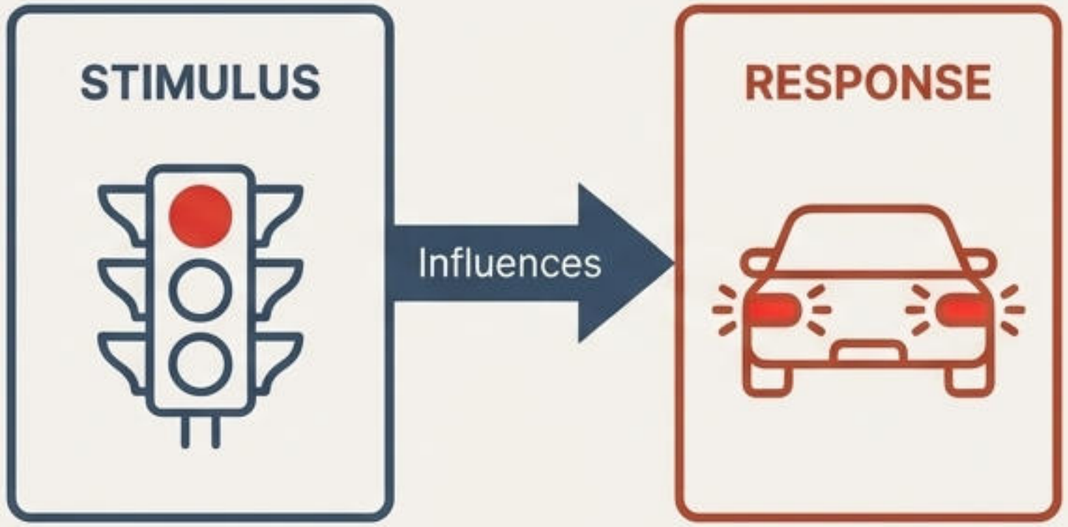
Stimulus and Response: The Core Interaction

Two of the most common terms in behavioral science are stimulus and response. A **stimulus** is any event that can potentially influence behavior. A **response** is a particular instance of a behavior that occurs as a result of a stimulus. The plural of stimulus is stimuli.

For example, food presented to a hungry dog is a stimulus that elicits the response of salivation. Loud music (a stimulus) might cause a neighbor to bang on the wall (a response).

A stimulus is sometimes referred to as a **cue** because it can act as a signal for a certain behavior, such as a red traffic light signaling the action of stopping.

Stimulus-Response Pathway



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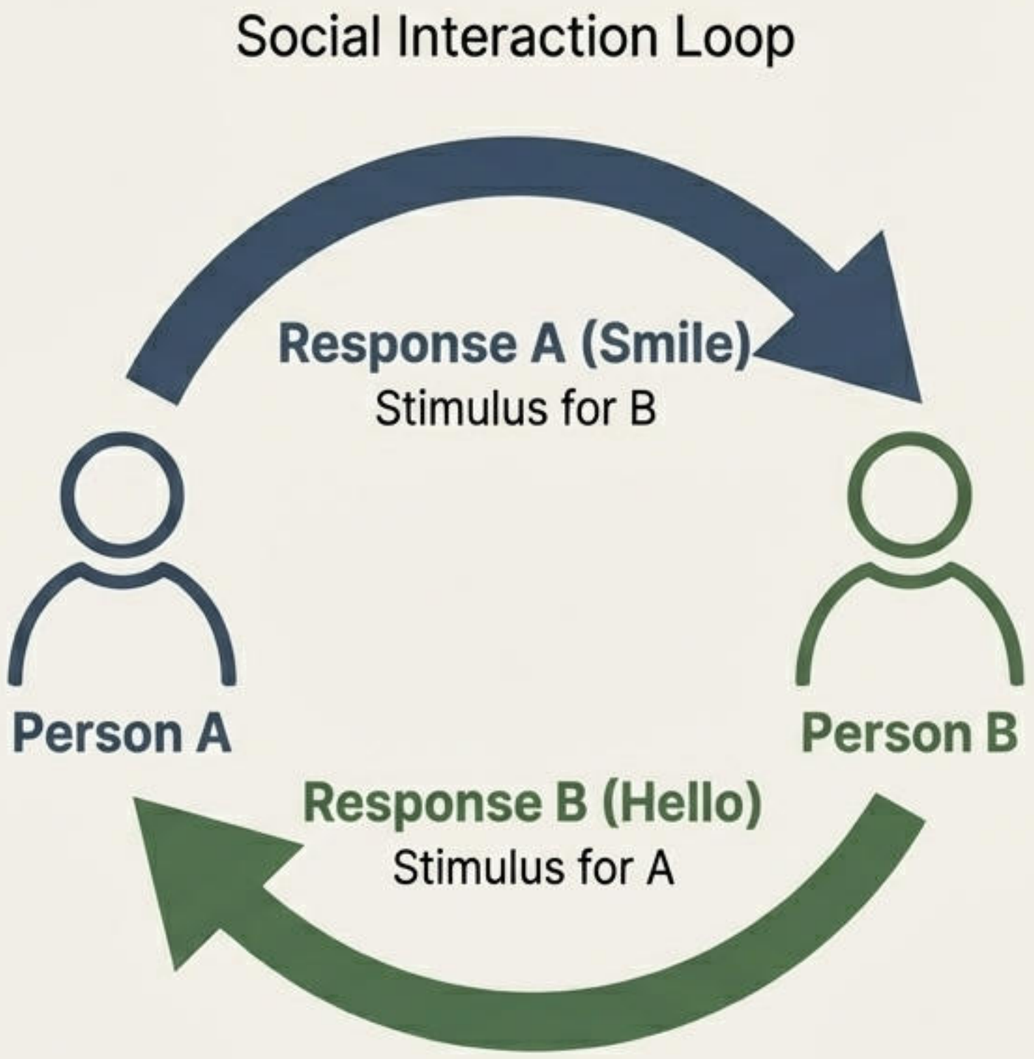
Stimulus-Response Chains in Social Interaction

The response of one organism can act as a stimulus that influences the response of another. This creates a chain of alternating responses that forms the basis of social interactions. Each person's behavior serves as a stimulus for the other person's next behavior. For instance, a smile from Shane is a stimulus that encourages Nav to respond by saying hello. Nav's hello, in turn, acts as a stimulus that encourages Shane to introduce himself. Similarly, if one rat bites another, the bite is a stimulus that might elicit a retaliatory response, which then acts as a stimulus for the first rat to retreat. Social dynamics can be understood as these continuous stimulus-response chains.



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Classifying Behavior: Overt vs. Covert

Behavior can be classified as either overt or covert. **Overt behavior** is behavior that can be directly observed by an individual other than the one performing it. It is publicly observable. A person saying hello or a rat pressing a lever are examples of overt behavior. Behaviorists have traditionally emphasized the study of overt behavior.

Covert behavior (or private events, as termed by B.F. Skinner) is behavior that can be perceived only by the person performing it. It is subjectively perceived and not publicly observable. Examples include dreaming, dreaming, thinking, visualizing, and feeling anxious. While some covert behaviors have overt components (e.g., anxiety can involve a measurable increase in heart rate), the internal experience itself remains private. Events both outside the skin (overt) and inside the skin (covert) can influence each other.

Overt Behavior

Directly observable by others.

- Saying hello
- Pressing a lever
- Walking

Covert Behavior

Perceived only by the performer.

- Thinking
- Feeling anxious
- Dreaming

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Classifying Stimuli: Appetitive vs. Aversive

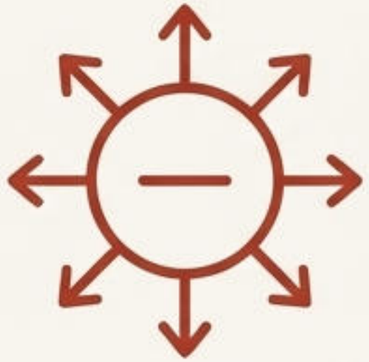
Stimuli can be classified based on how an organism interacts with them. An **appetitive stimulus** is an event that an organism will actively seek out. Examples include food for a hungry person or water for a thirsty one. These are often described as “pleasant.” An **aversive stimulus** is a event that an organism will avoid. Electric shock and extreme heat are examples of aversive stimuli. These are often described as “unpleasant.” Caution is necessary when relying on verbal descriptions, as a person's actions may contradict their words. For instance, a smoker may describe their habit as “disgusting” (aversive) but actively seek cigarettes (appetitive). The organism's behavior is the ultimate indicator of whether a stimulus is appetitive or aversive.

Appetitive Stimulus

An event an organism will seek out.

- Food (when hungry)
- Social contact

Aversive Stimulus



An event an organism will avoid.

- Electric shock
- Extreme heat

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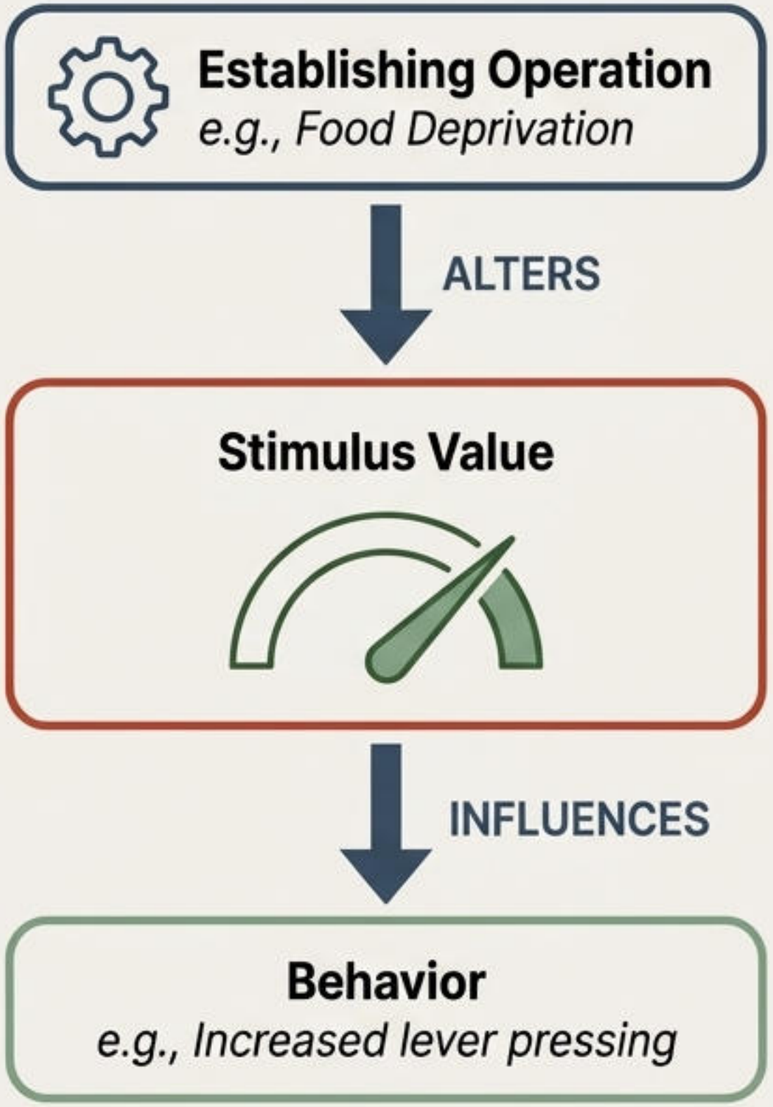
Establishing Operations: Modifying Stimulus Value

The appetitiveness or aversiveness of a stimulus is not fixed; it often depends on a particular state or condition. A procedure that affects or alters the appetitiveness or aversiveness of a stimulus is called an **establishing operation**. These operations establish a context in which a stimulus becomes more or less valuable to an organism, thereby influencing the behavior related to that stimulus. For example, food is highly appetitive to a hungry rat, but may not be to a rat that has just eaten. The state of hunger is established through a procedure that alters the value of food. Two common types of establishing operations are deprivation and satiation.

How Establishing Operations Work



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Types of Establishing Operations: Deprivation and Satiation

Deprivation is the prolonged absence of an event. This procedure typically increases the appetitiveness and reinforcing value of that event. For example, going without food for a long period (deprivation) increases the appetitiveness of food. Similarly, social deprivation can increase the desire for social contact.

Satiation is the prolonged exposure to or consumption of an event. This procedure often decreases the appetitiveness of that event. For example, food is a less effective reinforcer for a rat that is 'satiated' after a large meal. Listening to a favorite song too many times can lead to satiation, where the song loses its appeal. While these are general rules, exceptions like anorexia nervosa exist, where severe deprivation deprivation may paradoxically decrease the appetitive value of food.

Operation	Definition	Effect on Stimulus Value
Deprivation	Prolonged absence of an event.	Increases appetitiveness ▲
Satiation	Prolonged exposure to or consumption of an event.	Decreases appetitiveness ▼

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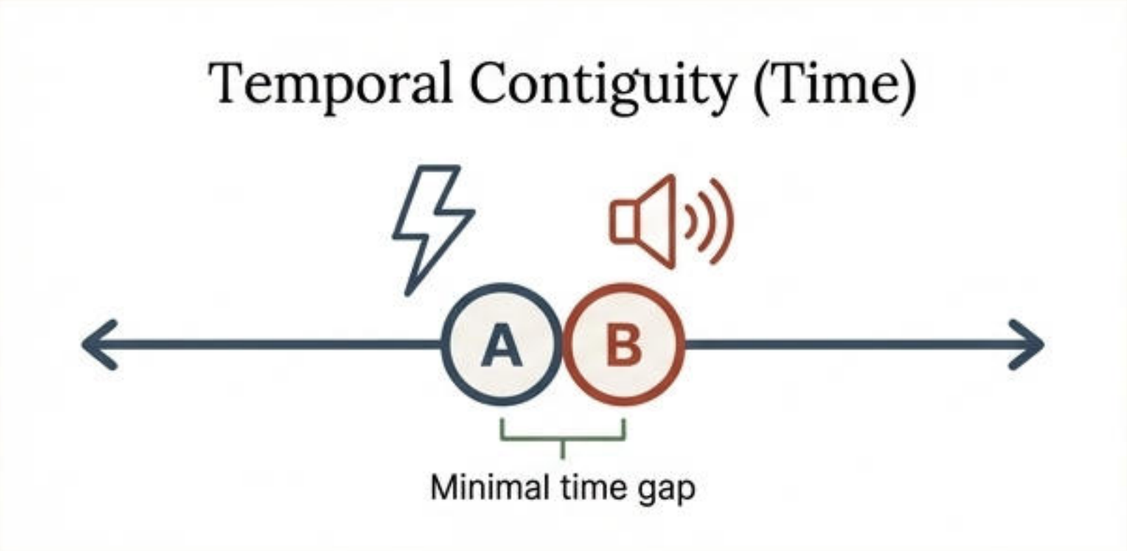
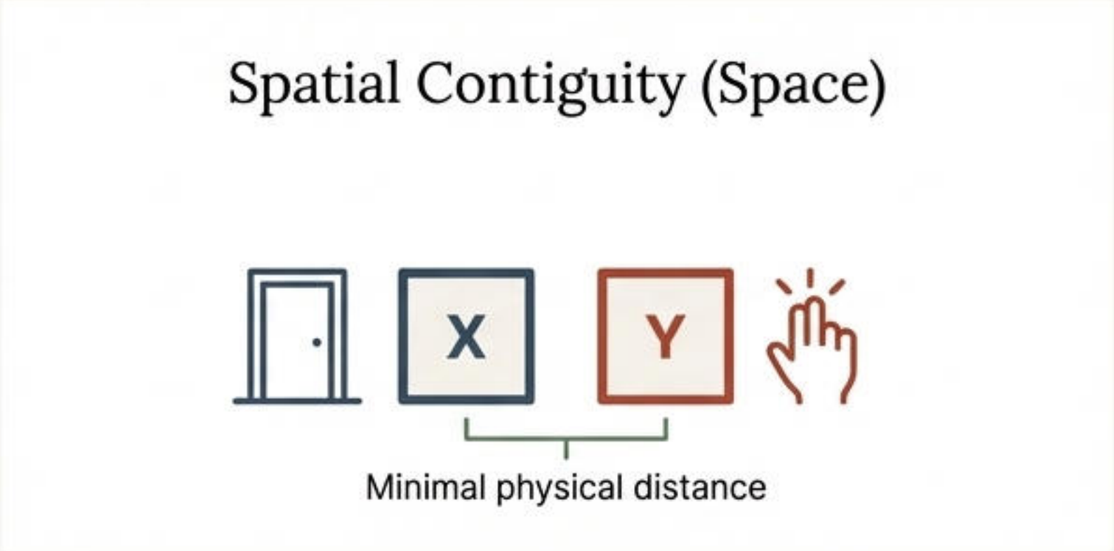
Contiguity: The Principle of Nearness

The concept of **contiguity** refers to closeness or nearness. It is an important factor in learning and has two primary forms: temporal and spatial. **Temporal contiguity** is the extent to which events occur close together in time. For example, lightning and thunder are temporally contiguous. A behavior is learned more readily when its consequence follows immediately. **Spatial contiguity** is the extent to which events are situated close to each other in space. A rat learns to press a lever for food more easily if the food dispenser is located **near the lever.** Similarly, a knock is spatially contiguous with a door, whereas a doorbell's sound often is not, which may make learning its meaning slightly more difficult.





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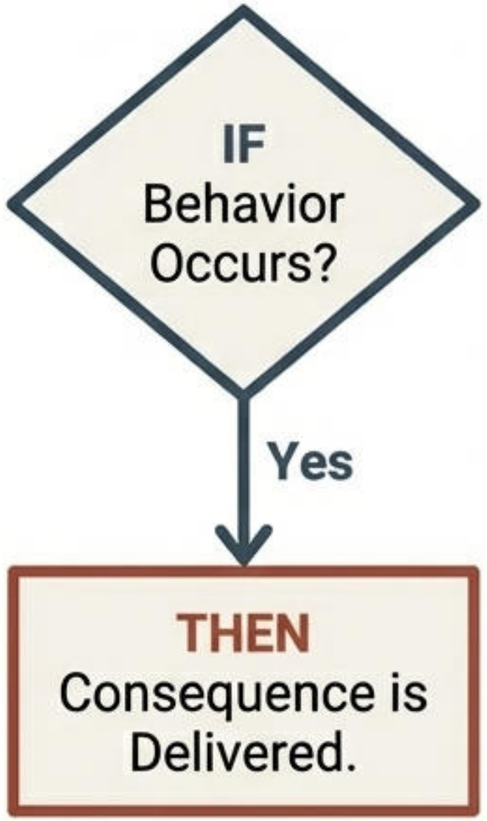
Contingency: The Predictive Relationship

Contingency refers to a predictive or functional relationship between two events, where the occurrence of one event predicts the probable occurrence of another. It is a different concept from contiguity. If a rat receives a food pellet *whenever* it preses a lever, a contingency exists between the behavior (lever pressing) and the consequence (food). We say that the food is *contingent on* the lever press. Likewise, if a child receives a balloon *every time* she goes to the dentist, a contingency exists between the visit and the balloon. The balloon is contingent upon visiting the dentist.

Contingency is a critical aspect of learning, as it establishes a reliable relationship between actions and their outcomes.

Contingency: The "If-Then" Rule



Example: **IF** a dog begs, **THEN** it receives a biscuit.

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Measurement Foundation: Behavioral Definitions

When studying behavior, it is critical to use proper **behavioral definitions**. These definitions must meet two criteria: they must be objective and unambiguous.

1. **Objective:** The definition must refer to an observable aspect of behavior. For example, defining aggression in terms of 'yelling and striking' is objective, whereas defining it by 'feelings of anger' is not, because feelings cannot be directly observed by others.
2. **Unambiguous:** The definition must be clear and precisely defined. For instance, 'yelling' could be defined as 'a loud vocalization that continues for more than 5 seconds and can be heard outside a closed door.'

Clear definitions ensure consistency in measurement, help verify that changes in behavior are due to the experimental treatment, and allow other researchers to replicate the study accurately.

Criteria for a Strong Definition

Objective
Refers to observable actions.

Unambiguous
Clearly and precisely defined.

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Dimensions of Behavior: Rate, Intensity, and Duration

Behavior can be measured in several ways. Three common dimensions are rate, intensity, and duration.

- Rate of Response: The frequency with which a response occurs in a certain period of time. This is suitable for brief responses with a clear start and finish (e.g., number of lever presses per hour). A cumulative recorder is a classic device for graphing the rate of behavior over time.
- Intensity: The force or magnitude of a behavior. This is measured when the strength of the response is important (e.g., the amount of saliva produced by a dog, the force of a rat's lever press).
- Duration: The length of time that an individual repeatedly or continuously performs a behavior. This is used when the goal is to increase or decrease the time spent on an activity (e.g., total time spent studying per day).

Rate

Frequency / Time

Intensity

Force or Magnitude

Duration

Total Time of Behavior

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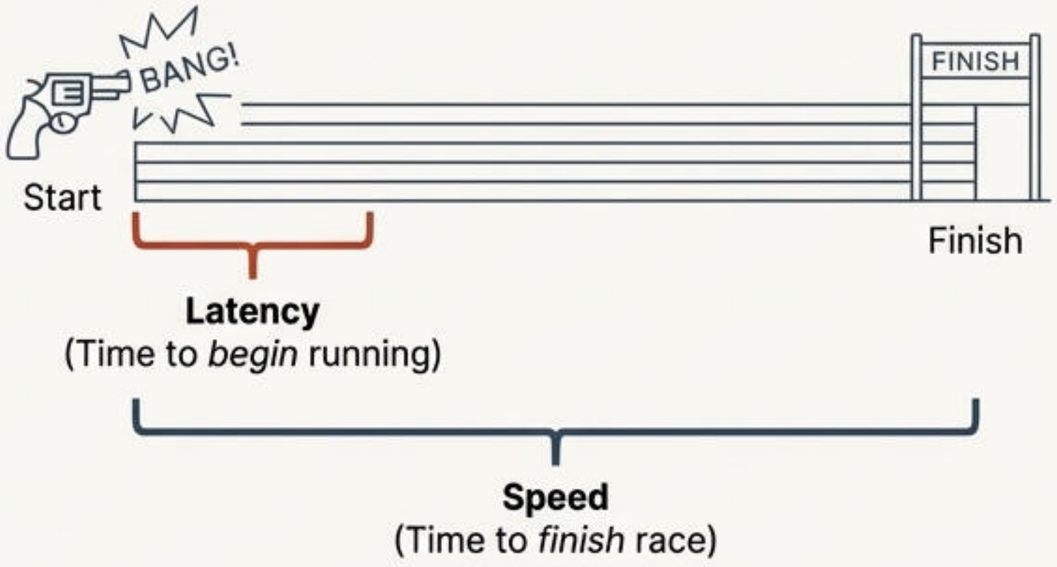
Temporal Dimensions of Behavior: Speed and Latency

Speed and latency are two additional measures related to time, but they are distinct from duration.

- **Speed:** A measure of how quickly or slowly a behavior occurs, or the rapidity of progressing through a distance. For example, the time it takes for a rat to run through a maze from the start to the goal box is a measure of speed.
- **Latency:** The length of time required for a behavior to *begin* after a specific event or cue. For example, in a conditioning experiment, latency could be how soon a dog begins salivating after hearing a tone.

Distinction Example (Sprinter):

- **Latency:** The time from the starting pistol until the runner begins moving.
- **Speed:** The time it takes to complete the 100-meter race.
- **Duration:** The total amount of time spent training each day.



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Observational Methods: Interval and Time-Sample Recording

Two efficient methods for measuring behavior, especially in applied settings, are interval and time-sample recording. In both methods, the focus is on whether a behavior occurred, not how many times.

- **Interval Recording:** The measurement of whether or not a behavior occurs during each interval within a series of *continuous* intervals. For example, an observer records if aggression occurs within each successive 10-minute block of a 3-hour period. The final measure is the percentage of intervals in which the behavior occurred.
- **Time-Sample Recording:** The measurement of whether or not a behavior occurs during each interval within a series of *discontinuous* (spaced-apart) intervals. For instance, an observer enters a classroom for a 10-minute interval at the start of each half-hour to record aggression. This method is highly time-efficient.

Interval Recording (Continuous)

Time-Sample Recording (Discontinuous)

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Qualitative Dimensions of Behavior: Topography and Errors

Finally, behavior can also be assessed based on its form and accuracy.

- **Topography:** The **actual physical form or structure** of a behavior. This is the focus when the quality of the movement is important, not just its occurrence. Examples include observing *how* a rat presses presses a lever (with which paw), teaching a child to write neatly, or training a dolphin to swim through a hoop.
- **Number of Errors:** A measure used for any behavior that can be categorized as right or wrong. The number of errors provides a direct assessment of performance or learning. Examples include counting the number of wrong turns a rat takes in a maze or the number of incorrect answers on a student's exam.

Topography

The physical form of a behavior.

Number of Errors

The count of incorrect responses.

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